

Course Title	Exploratory Data Analysis		
Course Code	IT462	Credit Structure	3-0-2-4
Course Format	It is 3 hours lecture and 2 hours Laboratory every week.		
Category	Program Elective	Semester	Autumn Semester (AY 2026-27)
Program	Both UG & PG		
Prerequisites (If any)	A strong foundation in basic probability & statistics, elementary calculus, Python packages.		
Brief Course Description	Exploratory Data Analysis (EDA) focuses on understanding data before formal modeling. The course teaches methods to inspect, summarize, clean, visualize, and transform datasets to discover patterns, anomalies, missingness, structure, and relationships. Students learn both classical statistical EDA and modern Python-based workflows for real-world data analysis.		
Course Objectives	The course's primary aim is to introduce students to the graphical and modeling techniques for exploring data, emphasizing visualization, interpretation, and clear communication of findings. The course will use modern software tools for data manipulation and visualization. It will build a foundation for exploring data using Python. The intention is that the student develops an intuition for statistical data analysis. The course emphasizes hands-on experiments with inbuilt as well as real datasets using the concepts/tools discussed in the theory classes. After completion of this course, students will be able to: • Understand data types, structures, and measurement scales. • Perform data cleaning and preprocessing. • Summarize and visualize univariate and multivariate data. • Detect missingness, outliers, anomalies, and bias. • Perform feature engineering and transformation. • Communicate insights using statistical reasoning and dashboards. Students must adhere to principles of academic integrity, ensuring that all work submitted is their own, properly attributed, and free from plagiarism or unauthorized collaboration. Cheating and plagiarism will not be tolerated in the tests, assignments, projects, etc. It will result in a Fail grade for the work or examination and may lead to disciplinary actions. Use of AI is encouraged as your assistant, however generating class assignments, project reports and any graded components will consider as plagiarism if not given credit for the way you use the AI.		

Evaluation Scheme	A: Lab Assignments / Quizzes 25% M: Two In-Semester Examinations 25% P: Final Project 50% The final score will be computed as: $S = 0.25 A + 0.25 M + 0.5 P$
Grading Policy	All make-up exams are scheduled and given at the instructor's discretion. Make-up exams are only given to those students who coordinate the missing of an exam before the originally scheduled exam date and time or for an emergency. Final letter grades (AA-F) will be assigned using relative grading based on class performance, final score distribution, minimum passing requirements, and instructor discretion.
Course Materials/References	Students will attend the instructions in class, participate in the discussions, submit quizzes, and complete several assignments. You will browse the course materials through Moodle portal. Textbook: 1. Exploratory Data Analysis by J W Tukey. Reference Books: 2. Visualizing Data by Cleveland 3. Think Stats: Exploratory Data Analysis by Allen B. Downey, 2014 4. Exploratory Data Analysis Using R by Ronald K. Pearson, CRC Press 5. Graphical Data Analysis with R by Unwin Antony 6. Graphical Exploratory Data Analysis by S. H. C. DuToit, A. G. W. Steyn, R. H.
Detailed Course Content	APPENDIX - I

Course Outcome:

CO1: Import, manipulate, clean, visualize, and export data in Python.

CO2: Conduct a systematic exploratory data analysis of different types of data.

CO3: Use research methods, including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

Program Outcomes (Mark it X if applicable refer Appendix – II)

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
	X		X	X							X

APPENDIX- I: Detailed Course Content (Session-wise/ Module-wise)

Module	Title & Description	# Lectures
Data and Statistics	What is Data? Data Science Pipeline, Role of EDA in ML, Structured vs Unstructured Data, Population vs Sample	3
Measurement Scales	Nominal Scale, Ordinal Scale, Interval Scale, Ratio Scale, Discrete vs Continuous Variables	4
Data Collection & Sampling	Census vs Sample, Sampling Bias, Random Sampling, Stratified Sampling, Cluster Sampling, Design of experiments	3
Data Import & Cleaning	Import data from different sources, Data Inspection, Getting insights about the dataset, Duplicate Detection, Inconsistent Formats, Invalid Entries, Data Type Conversion, Encoding Issues, Transformation	5
Missing Data Analysis	Missingness Mechanisms: MCAR, MAR, MNAR Missingness Visualization, Missing Data Treatment	5
Descriptive Summary	Measures of CT, Shape, Variation, Relationship...Mean, Median, Mode, Variance, Standard Deviation, IQR, MAD, Coefficient of Variation, Five number summary	3
Univariate & Bivariate Analysis	Numeric, Visualization methods: Histogram Density Plot, Boxplot, Violin Plot, Bar Chart, Scatter Plot, Covariance, Correlation, Crosstab, Heatmap	5
Correlation Analysis & Data fitting	Pearson Correlation, Spearman Correlation, Correlation Matrix, Multicollinearity, Regression models Linear & multiple regression, Goodness of fit	5
Advanced EDA, Building Dashboards	PCA, Time Series EDA, Trend Detection, Seasonality, ydata-profiling, Sweetviz, D-Tale Plotly Dashboards, Data Storytelling, Tableau, PowerB	5
Final Project	Collect your own data from data.gov.in Perform EDA on the collected dataset Discussion Discuss difficulties with peer/Instructor, Report submission, Presentation	2

An updated syllabus will be shared before the start of the class with the registered students.

Appendix – II : Program Outcomes (PO)

Engineering Graduates will be able to:

1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.